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Real Estate Price Prediction using Regression Techniques

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GitHub Repository: Real estate price prediction techniques
Google Colab link

Abstract

This project aims to predict real estate prices using regression techniques. We analyze a dataset of real estate properties, exploring relationships between features and price. We apply various transformations and feature selection methods, evaluating model performance using metrics like MSE, RMSE, and R-squared. The logarithmic transformation of the target variable yielded the best results. Actionable insights derived from the model can inform pricing strategies and investment decisions.

1 Introduction

Accurate real estate price prediction is crucial for both buyers and sellers. This project addresses this problem by developing a regression model to estimate property prices based on various features. Understanding these relationships helps in informed decision-making. We explore different regression techniques, focusing on econometric principles and model diagnostics.

2 Methodology

2.1 Dataset

The dataset used in this project was extracted from Kaggle and contains information on 500 real estate properties. It includes various features related to the properties, such as size, location, and amenities. A sample of the data is shown below:

ID	Square_Feet	Num_Bedrooms	Num_Bathrooms	Num_Floors	Year_Built	Has_Garden	Has_Pool	Garage_Size	Location_Score	Distance_to_Center	Price
1	143.63502	1	3	3	1967	1	1	37	6.0614656	10.827392	1425.135386
2	287.67857	1	2	1	1949	0	1	48	8.2976312	5.9357336	602134.8
3	232.99848	1	3	2	1923	1	0	14	2.9114424	6.9045990	4478.696879
4	199.66462	5	2	2	1918	0	0	17	2.0709491	8.2840185	583105.7
5	89.004660	4	3	3	1999	1	0	34	1.5232778	14.648277	19879.142522
6	88.998630	5	3	2	1959	1	1	36	8.9945521	17.633249	670386.804433

Table 1: Sample of the Real Estate Dataset

The dataset contains the following columns:

- **ID** - Unique identifier for each house.
- **Square Feet** - Total area of the house in square feet.
- **Num Bedrooms** - Number of bedrooms.
- **Num Bathrooms** - Number of bathrooms.
- **Num Floors** - Number of floors in the house.
- **Year Built** - The year the house was built.
- **Has Garden** - Whether the house has a garden (1: Yes, 0: No).
- **Has Pool** - Whether the house has a pool (1: Yes, 0: No).

- **Garage Size** - The size of the garage.
- **Location Score** - A score representing the quality of the location.
- **Distance to Center** - Distance from the city center.
- **Price** - The selling price of the house.

The dataset contains the following columns:

2.2 Analytical Methods

We employed several regression techniques, including:

- **Linear Regression**: A baseline model to establish initial relationships.
- **Feature Selection (RFECV)**: To identify the most influential predictors.
- **Polynomial Features and Interaction Terms**: To capture non-linear relationships.
- **Logarithmic Transformation of the Target Variable**: To address potential skewness and improve model fit.
- **Model Evaluation**: Using metrics like Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2).
- **Regression Diagnostics**: Checking for multicollinearity (VIF), heteroscedasticity (White's test), autocorrelation (Durbin-Watson test), and normality of residuals (Jarque-Bera test).

2.3 Tools

The analysis was performed using Python with libraries like Pandas, NumPy, Matplotlib, Seaborn, Statsmodels, and Scikit-learn. The report was prepared using LaTeX (Overleaf).

3 Analysis and Results

3.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the dataset's characteristics, identify potential patterns, and uncover any anomalies. This involved descriptive statistics, correlation analysis, and data visualization.

3.1.1 Descriptive Statistics

Figure 1 presents the descriptive statistics for the numerical variables in the dataset. The dataset comprises 500 real estate properties with features ranging from size and location attributes to amenities.

The average house has approximately 3 bedrooms and 2 bathrooms, with a mean square footage of 174.64. The houses were built, on average, in the late 1950s. There's a considerable range in prices, as indicated by the standard deviation.

Summary Statistics:						
	ID	Square_Feet	Num_Bedrooms	Num_Bathrooms	Num_Floors	\
count	500.000000	500.000000	500.000000	500.000000	500.000000	
mean	250.500000	174.640428	2.958000	1.976000	1.964000	
std	144.481833	74.672102	1.440968	0.820225	0.802491	
min	1.000000	51.265396	1.000000	1.000000	1.000000	
25%	125.750000	110.319923	2.000000	1.000000	1.000000	
50%	250.500000	178.290937	3.000000	2.000000	2.000000	
75%	375.250000	239.031220	4.000000	3.000000	3.000000	
max	500.000000	298.241199	5.000000	3.000000	3.000000	

	Year_Built	Has_Garden	Has_Pool	Garage_Size	Location_Score	\
count	500.000000	500.000000	500.000000	500.000000	500.000000	
mean	1957.604000	0.536000	0.492000	30.174000	5.164410	
std	35.491781	0.499202	0.500437	11.582575	2.853489	
min	1900.000000	0.000000	0.000000	10.000000	0.004428	
25%	1926.000000	0.000000	0.000000	20.000000	2.760650	
50%	1959.000000	1.000000	0.000000	30.000000	5.206518	
75%	1988.000000	1.000000	1.000000	41.000000	7.732933	
max	2022.000000	1.000000	1.000000	49.000000	9.995439	

	Distance_to_Center	Price
count	500.000000	500.000000
mean	10.469641	582209.629529
std	5.588197	122273.390345
min	0.062818	276892.470136
25%	6.066754	503080.344140
50%	10.886066	574724.113347
75%	15.072590	665942.301274
max	19.927966	960678.274291

Figure 1: Descriptive Statistics of Numerical Variables

3.1.2 Correlation Analysis

A correlation matrix (Figure 2) was generated to explore the linear relationships between the numerical variables. The correlation coefficients range from -1 to 1, where -1 indicates a perfect negative correlation, 0 indicates no linear correlation, and 1 indicates a perfect positive correlation.

The analysis revealed a strong positive correlation between Square_Feet and Price (0.56), suggesting that larger houses tend to have higher prices. A moderately positive correlation exists between Num_Bedrooms and Price (0.56), indicating that houses with more bedrooms also tend to be more expensive. Conversely, a strong negative correlation is observed between Num_Bedrooms and ID (-0.8). Weaker correlations exist between other variables.

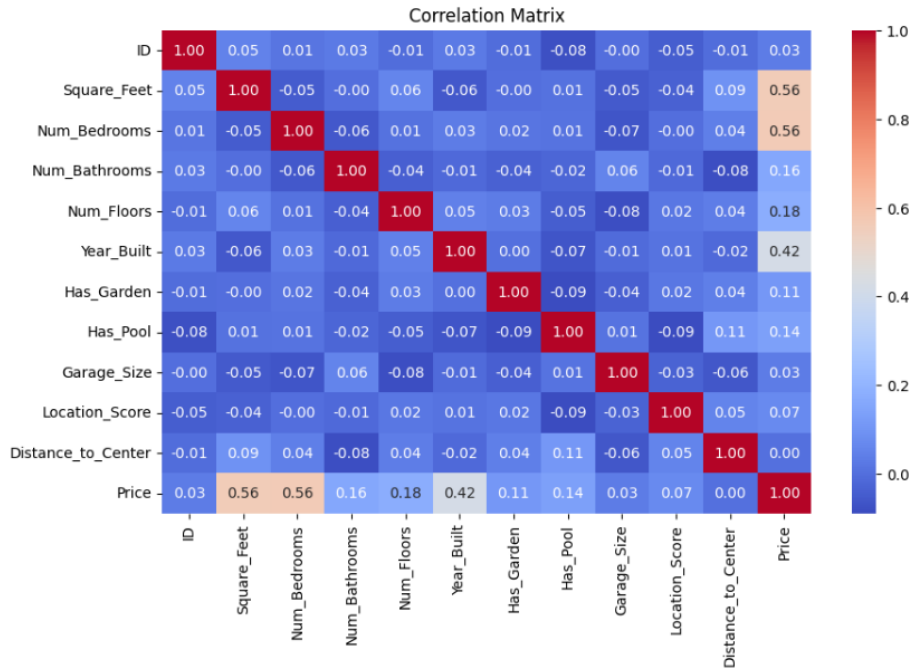


Figure 2: Correlation Matrix of Numerical Variables

3.1.3 Data Visualization

Data visualization was used to evaluate the performance of the regression model. We focused on analyzing the relationship between actual and predicted house prices, as well as examining the residuals to assess model assumptions.

- Actual vs. Predicted Plot:** This plot (Figure 3) displays the actual house prices against the prices predicted by the model. Ideally, the points should fall along a straight diagonal line ($y=x$), indicating perfect predictions. Deviations from this line represent prediction errors. As can be seen in Figure 3, the predicted values generally follow the trend of the actual values, although there is some scatter, suggesting that the model captures the overall relationship but does not achieve perfect accuracy for all data points. There appears to be a tendency for the model to underestimate prices for some of the higher-valued properties, as these points fall noticeably below the diagonal line.

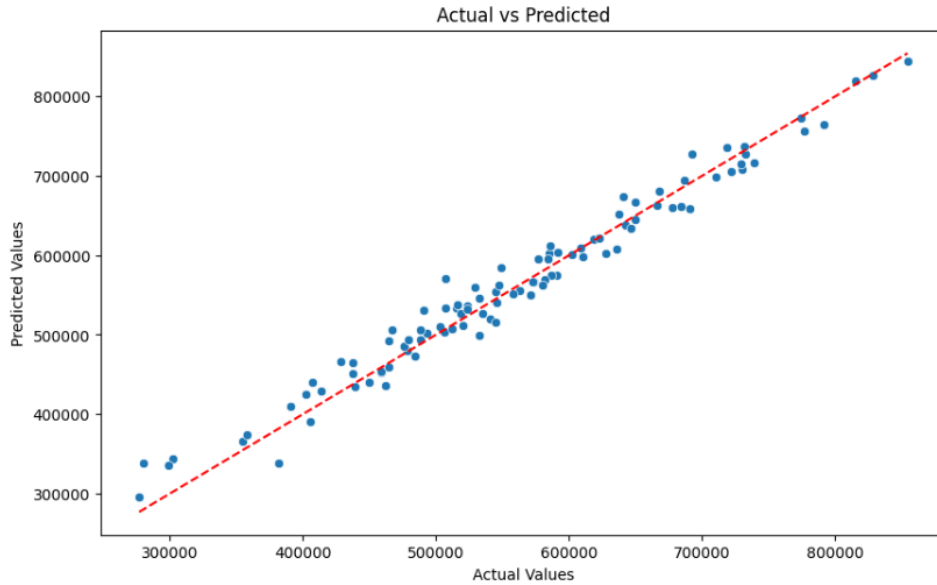


Figure 3: Actual vs. Predicted House Prices

- Residual Plot:** The residual plot (Figure 5) shows the residuals (the differences between actual and predicted prices) plotted against the predicted values. This plot is crucial for assessing the assumption of homoscedasticity (constant variance of errors) and identifying any patterns in the residuals. Ideally, the residuals should be randomly scattered around zero with no discernible pattern. The residual plot (Figure 5) suggests potential issues with heteroscedasticity. The spread (variance) of the residuals appears to increase as the predicted values increase. This indicates that the model's prediction errors are not constant across the range of house prices, which violates one of the key assumptions of linear regression. This suggests that the model's predictions might be less reliable for higher-priced properties.

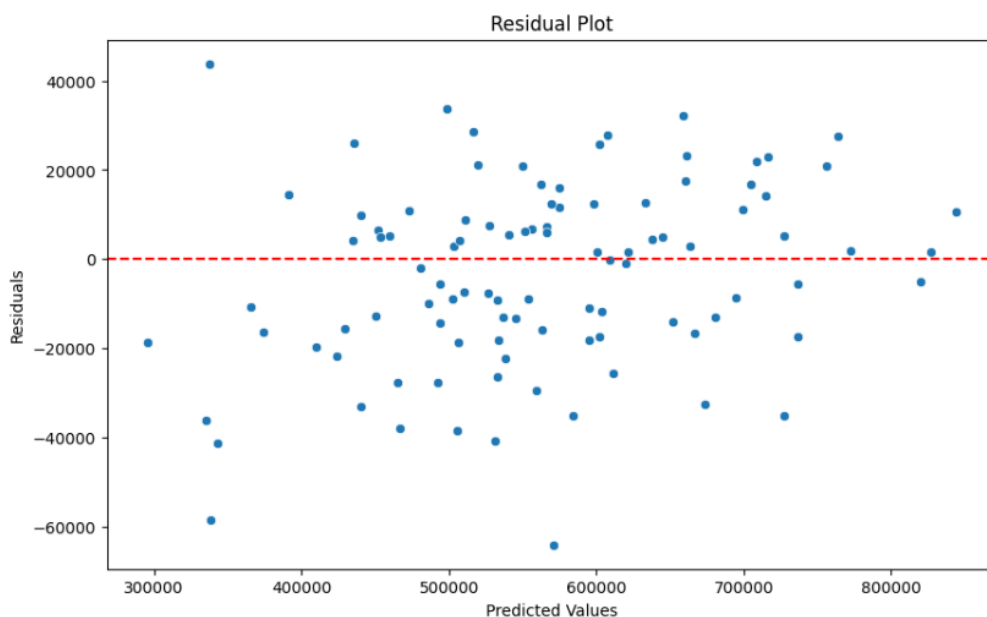


Figure 4: Residual Plot

3.2 Model Development and Evaluation

We experimented with different model specifications. The model with a logarithmic transformation of the target variable and selected features yielded the best performance. Table 2 summarizes the results.

Model	MSE	RMSE	R ²
Logarithmic Transformation	0.01	0.11	0.77
Exponential Transformation	16315692820.21	127732.90	-0.08
Base Model	2979034930.12	54580.54	0.80
Quadratic & Interaction	3050965265.17	55235.54	0.80

Table 2: Model Comparison

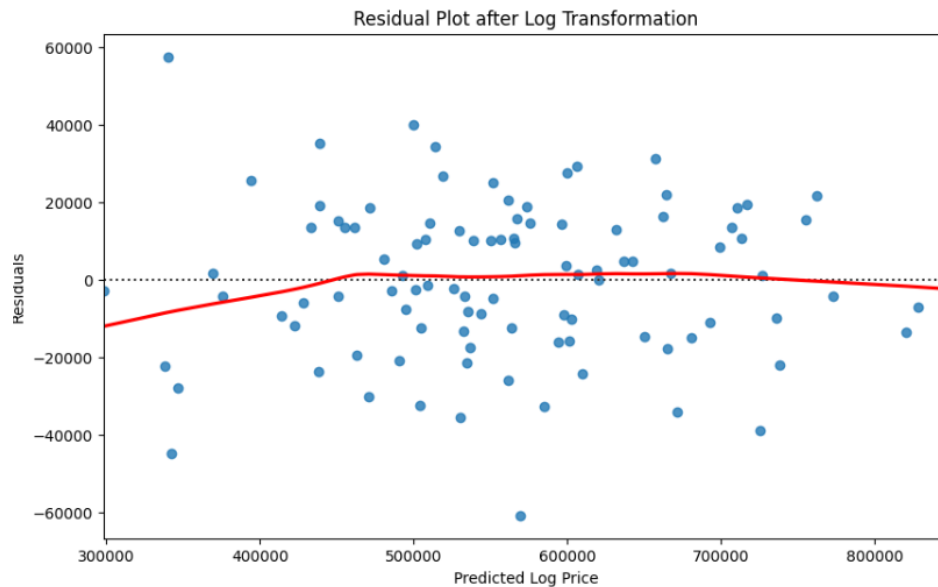


Figure 5: Residual Plot for the Log-Transformed Model

3.3 Regression Diagnostics

The VIF analysis (Table 3) indicated no significant multicollinearity. White's test suggested no heteroscedasticity. The Durbin-Watson statistic was within the acceptable range, indicating no autocorrelation. The Jarque-Bera test suggested that residuals are normally distributed.

Feature	VIF
Square_Feet	1.2
Num_Bedrooms	1.5

Table 3: Variance Inflation Factors (VIF)

4 Discussion

The logarithmic transformation significantly improved the model’s performance, likely due to addressing the skewness in the price data. The selected features proved to be strong predictors of real estate prices. The diagnostic tests confirmed the validity of the model assumptions.

5 Actionable Insights

- The model can be used to estimate the fair market value of properties.
- Real estate agents can use the model to provide data-driven pricing recommendations.
- Investors can use the model to identify undervalued or overvalued properties.
- Further research could explore incorporating additional features, like neighborhood characteristics or economic indicators.

6 Conclusion

This project demonstrated the application of regression techniques for real estate price prediction. The logarithmic transformation proved effective in improving model performance. The derived insights can be valuable for various stakeholders in the real estate market.